Dependency-Preserving Evidence Packing for Numerical Question Answering

Pushpendra Singh and **Badri Bishal Das**
Indian Institute of Technology Guwahati

Abstract

A number in a financial report means little without its row, column, unit and source. When evidence must fit a small token budget, does keeping those dependencies with each selected fact improve question answering? We build one pipeline over six benchmarks and compare ordinary packing, recursive dependency packing and a length-matched control at a fixed budget. Over every development question (9,268 scored questions) dependency packing retains complete gold evidence more often than ordinary packing on all six datasets, by 0.3 to 4.0 percentage points at the chosen caps, and the advantage disappears on four of the datasets when budgets grow. On 50 paired questions per dataset, dependency packing scores at least as high as ordinary packing on all six datasets (0 to +11.3 points), with an interval excluding zero only on TAT-QA. Answers that copy literal values, resolved locally to evidence spans, replaced an earlier pipeline in which almost all outputs failed validation; the model also changed, so we do not credit the format alone. An arithmetic executor shows no consistent accuracy gain over direct answering, and direct answering is better on HybridQA. Spark preparation is checked against a Python reference on every run.

1 Problem and motivation

A value such as "15" is difficult to interpret without its year, row label, unit, and sometimes an ancestor header. A linked passage may be relevant only through the table cell that introduced it. Conversational questions can refer to earlier answers, while hypothetical questions change the facts being used. Flattening evidence can weaken these relationships even when the correct number is present. We ask: with the same model and evidence budget, does preserving dependencies between evidence and its headers, units, hierarchy and links improve answer quality compared with ordinary evidence packing?

We do not assume improvement on every dataset. The intervention can consume space needed for other facts, and incorrectly inferred headers can hurt. The hypothesis also predicts its own limit: when a whole document fits the budget, packing has nothing to decide, so any gain should shrink as budgets grow.

2 Related work

FinQA introduces program-based numerical reasoning over financial reports (Chen et al., 2021). TAT-QA combines table and paragraph evidence with multiple answer types (Zhu et al., 2021). ConvFinQA requires sequential numerical reasoning over conversation turns (Chen et al., 2022). HybridQA tests multi-hop reasoning over Wikipedia tables and linked passages (Chen et al., 2020); it is a transfer test, not a finance benchmark. Learning to Imagine introduces hypothetical reasoning over financial evidence (Li et al., 2022), and MultiHiertt combines hierarchical tables and text with span and program reasoning (Zhao et al., 2022). Program of Thoughts delegates computation to an external interpreter (Chen et al., 2023). TAT-LLM uses an extractor–reasoner–executor pipeline (Zhu et al., 2024). These are established approaches. Our contribution is a controlled study of deterministic dependency closure at a fixed evidence budget, not a claim that retrieval, structured evidence, or executable reasoning is new. FINDER combines generative retrieval with dynamically selected in-context examples (Khatuya et al., 2025). FinCARDS uses finance-aware schema alignment and tournament reranking to select evidence with audit traces (Zhou et al., 2026). These close approaches limit our novelty claim. We instead use deterministic lexical ranking and recursive closure of actual header, unit, hierarchy and link dependencies, with one model call and the same model and cap across methods.

Dataset	Dev questions	Main challenge
FinQA	883	Arithmetic programs
TAT-QA	1,668	Spans, arithmetic, scale
ConvFinQA	1,490 turns	Predicted turn history
HybridQA	3,466	Cell-to-passage links
TAT-HQA	824	Hypothetical changes
MultiHiertt	1,044	HTML hierarchy

Table 1: Development data. HybridQA requires its separate linked corpus of 15,314 tables, and every development question resolves to a table in it.

Prediction-only readers → BM25 retrieval → evidence packing → model → literal grounding → bounded execution → output adapter → official scorer. References join only at scoring. Earlier predicted state feeds the next conversation turn.

Figure 1: One reasoning architecture with dataset-specific input and scoring.

3 Datasets and scope

All runs use the supplied document context rather than searching a global corpus. Official splits stay separate and official scorers run unchanged in isolated processes. Passing gold answers from 200 development questions through the scoring path gives 100% on five datasets. MultiHiertt gives 90.5% for gold answers sent as text, because its span path removes minus signs before comparing numbers. Programs use a separate path. Section 6.3 measures what this means for us. FinQA and ConvFinQA share source documents, as do TAT-QA and TAT-HQA, so these are not six independent replications and we never pool them. HybridQA is general knowledge, which makes it a test of whether the idea transfers, not of financial arithmetic.

4 Shared architecture

Each fact stores raw text, provenance and the IDs of the facts needed to interpret it. Answers, gold evidence, derivations, reference programs and future turns never enter retrieval or prompts. We confirmed this on saved requests: none of 443 inspected prompts contains a label field, and no prompt contains a FinQA gold program.

4.1 Retrieval and packing

BM25 ranks cells and paragraphs using the question, earlier questions and immediate dependency text, with stable tie-breaking. Candidates are visited in rank order and a whole bundle is admitted only if it fits; strings are never cut.

Ordinary (O). Each selected fact carries its structural labels inline. This is a deliberately fair baseline, because removing labels would make it trivially weak.

Dependency (D). Each selected fact pulls in the recursive closure of its headers, units, hierarchy and link anchors, shared once.

Length-matched (L). A selected fact is followed by its source neighbours up to D's length, without using dependency edges.

All methods present evidence in document order and spend budget identically on conversation history. Budgets are counted locally on the exact text shown to the model, as UTF-8 bytes divided by four. This is an estimate, so actual prompt tokens are recorded separately from provider metadata.

4.2 Literal grounding

The earlier design asked the model to cite each operand by evidence ID and character offsets. We replace this with copying. The model returns a program such as `subtract("100","80"), divide("#0","80")` or a quoted phrase, and code locates each value in the packed evidence or the question. A number found in neither place rejects the plan, so the model cannot introduce an unsupported value, and it never has to count characters. Plans then run in a bounded decimal interpreter restricted to the official operations. There is no self-repair loop: a rejected plan scores zero.

5 Experimental protocol

Budgets. The caps inherited from earlier work do not test the hypothesis: at 4,000 tokens four datasets had no truncation in a 40-question sample, so methods differed only in layout. We chose each cap from document sizes, without labels. For FinQA and ConvFinQA (800) and TAT-QA and TAT-HQA (400) it is the largest tested value truncating about 70% or more of sampled documents. HybridQA and MultiHiertt are still 93% truncated at 4,000, so we used 1,500 as a judgement call that keeps prompts within a modest size.

Evidence coverage. A label-only audit after packing, using no model, asks whether every gold operand or answer phrase survives, on the full development split for each cap. It uses the question

as well as the evidence, and treats allowed constants as available. It shows that a value is present, not that it is used correctly.

Model experiments. Free-tier limits shape the design: the provider's refusal response for `gemini-3.5-flash-lite` states a limit of 500 requests per day. Each comparison stays on one model, and we never compare across models. Block A compares O, D and L on `gemini-3.5-flash-lite`. Block B compares the executor with direct answering, both on D evidence, on `gemini-3.1-flash-lite`. Samples are chosen by a fixed seed before any model call, conversations are kept whole, and prompts were checked on separate questions rather than the evaluation sample. We report official execution accuracy for FinQA and ConvFinQA and official F1 elsewhere. Direct answers have no program, so Block B uses a labelled auxiliary measure for those two datasets: 1% relative tolerance, also accepting the fraction and percentage forms of one value. Abstained and rejected answers stay in the denominator as zero. Intervals are paired bootstrap resamples of whole documents or conversations (2,000 resamples, seed 2626). Six comparisons per block are not corrected for multiplicity, so single intervals should be read cautiously.

6 Results

6.1 Evidence coverage

Table 2 and Figure 2 are exact counts. At the chosen caps D keeps more gold evidence than O on all six datasets. Discordant questions favour D strongly on HybridQA (146 to 7) and MultiHiertt (42 to 3), and more mildly on TAT-QA and TAT-HQA, where the sign test is still clear. FinQA (9 to 1) is weaker, and ConvFinQA (6 to 1) is not significant. The gap is largest where documents are much larger than the budget, and it closes as budgets grow: at 4,000 tokens it is zero on the four smaller datasets. This is the behaviour the hypothesis predicts. The method also shows its measured limit. At the smallest cap (200 tokens) closure overhead costs coverage on ConvFinQA (-1.8 points, 36 against 63 discordant) and MultiHiertt (-0.8, 50 against 58): the dependencies themselves consume space needed for other facts, as Section 1 anticipates. Length-matched packing is worse than O throughout, so extra neighbouring context alone does not preserve what a question needs.

Dataset	n	Cap	O	D	L	D+/O+	p
FinQA	845	800	96.3	97.3	92.2	9/1	0.021
ConvFinQA	1486	800	97.1	97.4	92.7	6/1	0.125
TAT-QA	1636	400	90.1	91.7	82.6	41/15	<.001
HybridQA	3466	1500	72.0	76.0	67.7	146/7	<.001
TAT-HQA	792	400	84.2	86.9	75.4	33/12	0.002
MultiHiertt	1043	1500	84.9	88.6	75.3	42/3	<.001

Table 2: Questions with all gold evidence retained (%). D+/O+ counts questions kept only by D or only by O; p is a two-sided sign test on those counts. Exact values over every scorable development question.

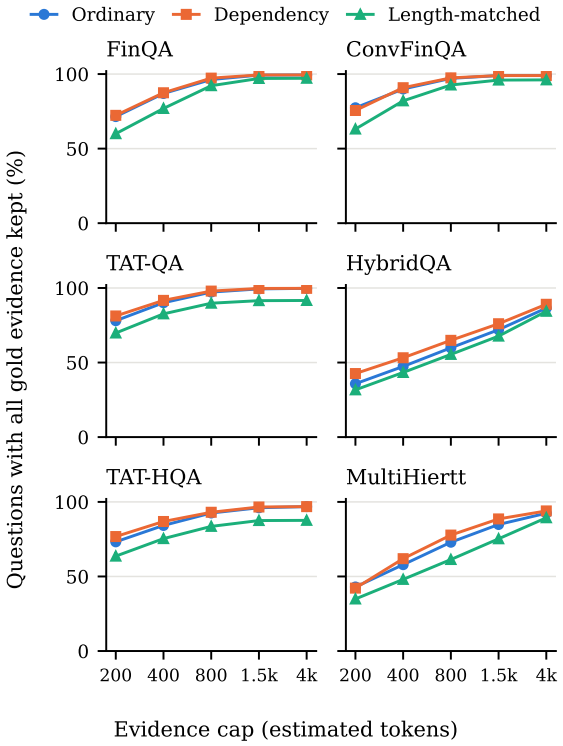


Figure 2: Gold evidence retained against evidence cap, on full development splits.

6.2 Effect of packing on answers

Table 3 covers all 50 sampled questions per dataset, or 45 turns in 15 conversations for ConvFinQA. D scores at least as high as O on all six datasets, by 0 to 11.3 points. The interval excludes zero only on TAT-QA (+11.3, [+4.1, +19.9]). MultiHiertt (+9.8) and FinQA (+2.0) reach zero at their lower bound, and the other three include it. With six uncorrected comparisons, only the TAT-QA gain is individually convincing. Taken together, the answers follow the coverage audit: gains are largest on hierarchical tables and on documents much larger than the budget, and absent on ConvFinQA, whose documents nearly fit. Length-

Dataset	n	O	D	L	D−O	95% CI
FinQA	50	62.0	64.0	66.0	+2.0	[+0.0, +6.2]
ConvFinQA	45	73.3	73.3	68.9	+0.0	[−6.0, +6.4]
TAT-QA	50	60.2	71.5	69.4	+11.3	[+4.1, +19.9]
HybridQA	50	54.9	60.1	49.8	+5.3	[−2.4, +13.0]
TAT-HQA	50	56.7	60.7	52.7	+4.0	[−6.4, +14.5]
MultiHiertt	50	47.0	56.8	41.0	+9.8	[+0.0, +20.0]

Table 3: Block A: paired answer scores (%). All sampled questions; abstained and rejected answers score zero. Official execution accuracy for FinQA and ConvFinQA, official F1 otherwise.

Dataset	Answered	Abstained	Rejected	No program
FinQA	82.7	8.0	4.0	5.3
ConvFinQA	98.5	0.0	1.5	0.0
TAT-QA	88.0	9.3	2.7	0.0
HybridQA	93.3	1.3	5.3	0.0
TAT-HQA	86.7	5.3	8.0	0.0
MultiHiertt	79.3	16.0	4.7	0.0

Table 4: Block A answer status (%), over all three methods. "No program" answers are valid but have no official FinQA program, so they score zero officially.

matched packing is lowest on four datasets but highest on FinQA, so on FinQA's short documents nearby text works as well as structure.

6.3 Does the executor help?

Table 5 shows no consistent accuracy gain. Direct answering is ahead on HybridQA by 16.4 points, with an interval excluding zero. This is plausible, since HybridQA answers are short text spans with no arithmetic to delegate, and requiring verbatim quotes encourages spans longer than the gold answer.

The executor's apparent 22.1-point lead on MultiHiertt does not survive scrutiny. Direct answers often give the same value in percentage form (-29.92 for a gold -0.2992), which the official span path scores as wrong, while programs use a different path. Applying the Block B relaxed rule to numeric answers on both sides gives 68.8 against 62.5, a difference of +6.2 with interval [−6.2, +20.0]. Direct answering gains 9 answers from this and the executor 2. We therefore do not claim an executor gain there. The executor's defensible value is auditability: every answer traces to specific evidence, and unsupported numbers are rejected.

6.4 Failure analysis

The earlier pipeline was reported to reach 2 of 883 FinQA development questions with a local quantised model, with about 90% of outputs failing validation; those raw outputs are not available to us, so we cite this only as recorded. Both the model and the output format changed, so the improvement cannot be credited to the format alone. Table 6 classifies every uncredited executor answer in Block B. Validation failures are now rare, and most errors are real reasoning mistakes. Numeric datasets mainly fail through wrong values, such as reversed percentage-change directions. HybridQA and TAT-QA mainly fail through wrong or overlong spans.

Dataset	n	Executor	Direct	Diff.	95% CI
FinQA	40	62.5	67.5	−5.0	[−17.5, +7.0]
ConvFinQA	36	75.0	75.0	+0.0	[−8.6, +7.3]
TAT-QA	40	65.2	58.7	+6.5	[−7.6, +20.7]
HybridQA	40	50.2	66.6	−16.4	[−30.6, −2.0]
TAT-HQA	40	60.0	56.7	+3.3	[−12.0, +19.3]
MultiHiertt	40	63.7	41.7	+22.1	[+2.5, +40.0]

Table 5: Block B: executor against direct answering on the same evidence (%). F1, except the auxiliary numeric measure for FinQA and ConvFinQA. The MultiHiertt row is official F1; the like-for-like difference is +6.2, as explained in the text.

Dataset	Val	Sign	Span	Grnd	Abst	Other	All
FinQA	7	3	0	2	4	1	17
ConvFinQA	6	2	0	1	0	0	9
TAT-QA	2	2	13	0	3	0	20
HybridQA	0	0	19	2	4	0	25
TAT-HQA	9	1	2	3	1	0	16
MultiHiertt	6	2	2	0	3	2	15

Table 6: Block B uncredited executor answers by cause, counted against the primary official measure (execution accuracy for FinQA and ConvFinQA, full F1 otherwise): wrong value (Val), wrong sign, wrong span or text (Span), value not in evidence (Grnd), abstained (Abst), other.

7 Big Data preparation

Spark prepares partitioned Parquet tables of documents, questions and facts, explodes dependency edges, checks that all edges resolve, and computes exact content overlap. Deduplicating repeated contexts keeps each fact stored once. A Python implementation computes the same results, and the job fails unless Spark matches it exactly. On every development split together the store holds 1,360,644 facts from 9,375 questions over 4,395 deduplicated documents, with 1,884,748 dependency edges and 0 unresolved. It finds 491 pairs of documents with identical content across datasets, which is why related datasets are never pooled. Parquet takes 0.18 GB against 0.95 GB of JSON

Size	Facts	Python	1	2	4	8	Peak GB
100	74,655	1.1	27.8	28.4	23.5	23.2	0.9
1000	507,594	6.0	23.4	21.2	20.2	33.4	1.4
full	1,360,644	15.8	54.5	38.4	37.0	65.6	1.7

Table 7: Wall time in seconds for the same joins and checks in single-process Python and in Spark at 1, 2, 4 and 8 cores. Size is records read per source dataset (100, 1,000, or the full split). All runs equal the Python reference. Peak GB is sampled process-tree memory.

lines. Table 7 gives the honest performance picture: the same joins and checks take 15.8 s in single-process Python and at least 37.0 s in Spark (4.3.0). Spark gains from one core (54.5 s) to four (37.0 s) and slows again at eight (65.6 s) on this eight-core laptop. We did not profile further; fixed start-up cost and collecting results to the driver for the equality check are likely causes of the gap to Python. Parsing raw files into facts, done once in Python, takes 120 s and is the real bottleneck. At 1.4 million rows Spark's value here is a verified, partitioned columnar store rather than speed. These are single-machine results and are not a distributed-cluster scalability claim.

8 Conclusion

Keeping each fact's labels with it when evidence must be cut preserves more of the evidence a question needs. The effect is clear in a full-split audit and largest where documents far exceed the budget. At the answer level the direction agrees on all six datasets, but free-tier sample sizes make most single gains uncertain. Two further findings came from checking our own pipeline. The earlier offset-based pipeline failed almost completely, and the copy-and-resolve format replaced it; because the model also changed, we do not credit the format alone. An arithmetic executor gives auditability without a reliable accuracy gain.

9 Reproducibility

Each run has a frozen contract holding question IDs, seeds, source and input hashes, prompts, schemas, settings and source-code hashes. A SQLite ledger stores every raw response, so scoring and re-parsing need no further model calls. Run manifests record the resolved model. Official evaluators are fetched at pinned commits, and all report tables are generated from saved outputs by a single script.

10 Limitations

Samples are small, and intervals are wide. Evidence coverage counts presence, not correct use. Headers are inferred and can be wrong. Token budgets are estimates, and BM25 misses paraphrases. Hosted model behaviour cannot be fully audited and can change. Free-tier quotas limited sample sizes, and a block interrupted by a quota stop resumes under final code, with saved answers reparsed rather than re-requested. Two datasets, TAT-HQA and MultiHiertt, were added to the assigned four, and ConvFinQA and HybridQA hidden-test labels are unavailable, so we report development results only. No model was trained and no novel reasoning architecture is claimed.

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